Level 3 Numeracy, Week 2: Multiplication, a Story and a Riddle

Why multiplication works, how to do it on paper, then a long story problem and a number riddle

FOR VOLUNTEERS

- This lesson may need **two days**. Part 1 is the idea of 2-digit × 1-digit. Part 2 is 2-digit × 2-digit and the standard method. Decide using the class response, the weakest child present, and the time you have.
- Make sure **every child** can do Activity 1 (place value) before moving on.
- At the end of Part 1, give a few more problems of the form **2-digit × 1-digit**. At the end of Part 2, show 2–3 more with sticks, and **never go beyond 2-digit × 2-digit**.

Part 1: the idea of multiplication

Activity 1: place value and money

Ask for the place value of each digit of these numbers: 12, 25, 56, 156, 2567, 84561.

Then use money. "I can give you Rs 25 as two ten-rupee notes and five one-rupee coins." Fill the table together:

Rupees	100-rupee notes	10-rupee notes	1-rupee coins
12	0	1	2
25	0	2	5
256	2	5	6

ANSWER KEY (VOLUNTEERS ONLY)

Place values: 12 = 10 + 2; 156 = 100 + 50 + 6; 2567 = 2000 + 500 + 60 + 7; 84561 = 80000 + 4000 + 500 + 60 + 1.

Activity 2: what does × mean?

One digit × one digit. Show 4 boxes with 3 kittens in each box. Count: 3 + 3 + 3 + 3 = 12. Instead of adding the same number again and again, we multiply:

4 boxes × 3 kittens in each box = 12 kittens

NOTE
To multiply quickly you must know the times tables. Practise one table at the start of every Level 3 class, for two minutes.

Two digits × one digit: 25 × 3. Show 3 groups, each made of 2 ten-rupee notes and 5 coins. Count: tens: 6 notes = 60. Ones: 15 coins = 15. Total 60 + 15 = 75.

KEY IDEA
Multiply the tens, multiply the ones, then add. 25 × 3 = (20 × 3) + (5 × 3) = 60 + 15 = 75.

Now the written method: ones first, carry the 1 ten to the tens column.

Step	25 × 3
Ones: 5 × 3 = 15. Write 5, carry 1.	15
Tens: 2 × 3 = 6, plus the carried 1 = 7.	7
Answer	**75**

ANSWER KEY (VOLUNTEERS ONLY)
Practice: 35 × 8 = 280; 56 × 6 = 336; 109 × 9 = 981.

Part 2: two digits × two digits

Activity 1: the pumpkin game

There are 12 sacks, with 14 pumpkins in each sack. How many pumpkins in all? Adding 14 twelve times is slow, so we write 14 × 12. Now we find the answer with sticks.

1. Lay out **1 stick and 4 sticks** (for 14) in one direction, side by side, with a small gap.
2. Lay out **1 stick and 2 sticks** (for 12) across the first set, so the two sets cross.
3. Count the crossing points **in three zones**: left-down (hundreds), middle (tens), right-up (ones).
4. Multiply each count by its place value and add: 1 × 100 + 6 × 10 + 8 × 1 = 100 + 60 + 8 = **168**.

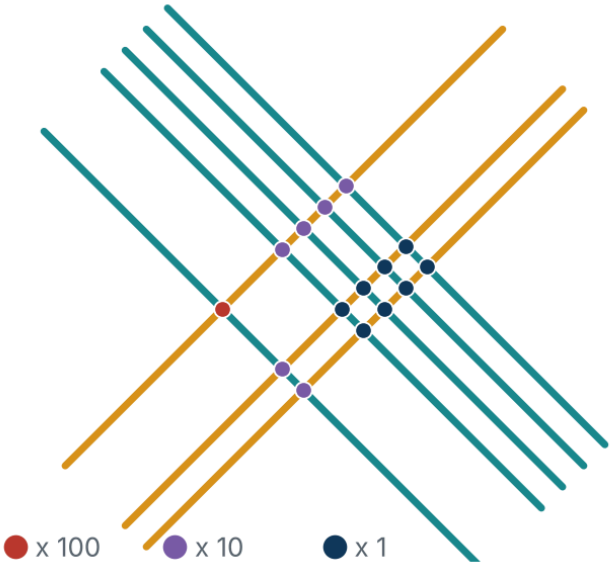


Figure 1. 14 × 12 with sticks. Count the dots in each coloured zone: ×100, ×10 and ×1.

Zone	Crossing points	Place value
Hundreds	1	1 × 100 = 100
Tens	6	6 × 10 = 60
Ones	8	8 × 1 = 8
Total		**168**

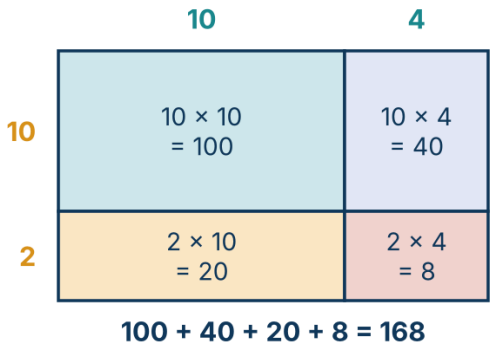


Figure 2. The same product as an area: (10 + 4) × (10 + 2) = 100 + 20 + 40 + 8 = 168.

TRY IT
If we swap the numbers, 12 × 14, do the sticks change? (The picture turns, but the number of crossing points stays the same: 14 × 12 = 12 × 14.) Practise with 15 × 11, 21 × 21, 10 × 12.

ANSWER KEY (VOLUNTEERS ONLY)
15 × 11 = 165; 21 × 21 = 441; 10 × 12 = 120.

Activity 2: the standard method

Write 14 as 10 + 4. Then 14 × 12 = (10 × 12) + (4 × 12).

- Multiplying by 10 means putting a **0** at the end: 10 × 12 = **120**.
- 4 × 12 = **48** (two digits × one digit, learned in Part 1).
- 120 + 48 = **168**, the same as the sticks gave.

Written in a column	
14 × 12 ——————— 28 (14 × 2) 140 (14 × 10) ——————— 168	First multiply by the ones digit (2), then by the tens digit (1 ten, so we put a 0 and write 140). Add the two lines.

TIP
Show a carrying example too, such as 27 × 13. 27 × 3 = 81 and 27 × 10 = 270, and 81 + 270 = 351. Ask: "Which numbers match the sticks method?" The four pieces 20 × 10, 7 × 10, 20 × 3, 7 × 3 give 200 + 70 + 60 + 21 = 351.

ANSWER KEY (VOLUNTEERS ONLY)
15 × 22 = 330 (10 × 22 = 220, and 5 × 22 = 110).

NOTE
In daily life we usually multiply big numbers on a calculator. What matters is to know why the method works, and to do easy multiplications in the head or on paper.

The story: Satyanweshi and the temple

A boy named Satyanweshi lived with his father Isbara in the small village of Bisba. Sometimes they walked to a famous temple nearby to pray. (The original spelt both names in several ways; we use one spelling each.)

1. The temple is **7 km** uphill, and it takes **1500 steps** to walk 1 km. How many steps from the bottom of the hill to the temple?
2. On the way, a neighbour in a hurry gives Satyanweshi **19 bananas** to share equally among **5 beggars**, and to keep whatever is left. How many bananas does he keep?

3. After walking **6000 steps**, Satyanweshi sees a dog being chased away by some boys with stones. How many steps are still left to the temple?
4. Coming down, they walk **half the way at 1500 steps per km** and **half at 2000 steps per km**. How many steps in total from the temple to the bottom?

ANSWER KEY (VOLUNTEERS ONLY)

- 1: 7 × 1500 = **10 500** steps.
- 2: 19 ÷ 5 = 3, remainder **4**. He keeps **4** bananas (each beggar gets 3).
- 3: 10 500 − 6000 = **4500** steps.
- 4: half of 7 km = 3.5 km. 3.5 × 1500 = 5250. 3.5 × 2000 = 7000. Total **12 250** steps. (If children find 3.5 hard, use "3 km and half a km": 3 × 1500 + 750 = 5250; 3 × 2000 + 1000 = 7000.)

FOR VOLUNTEERS

The original story has two optional moments for talk: why some people are kind and others are not, and why some people suffer more than others. These are big questions. Let children share if they want, accept all answers kindly, and do not give a religious or a final answer yourself. If the class is not ready, skip them; the maths does not depend on them.

The riddle: the temple treasure

One day Satyanweshi takes his friend Rahul to the old temple. In a big hall they find a locked chest with a 3-digit code and a riddle beside it:

THE RIDDLE

"I am a three-digit number. My first digit is an odd number more than 2. My second digit is the largest even number between 1 and 10. My third digit is greater than half of the second one and it is an even number. The sum of my three digits is 17. What number am I?"

ANSWER KEY (VOLUNTEERS ONLY)

- Second digit: even numbers between 1 and 10 are 2, 4, 6, 8, so the largest is **8**.
- Third digit: more than half of 8 (4), and even: **6** or **8**.
- The sum is 17, so first digit = 17 − 8 − third. If the third is 6, the first is **3** (odd, more than 2 ✓). If the third is 8, the first is 1, which is **not** more than 2 ✗.
- The code is **386**.

Only after the children solve the riddle, give them the next page. The chest opens: 140 gold coins and 80 diamonds. They decide to give each of 5 beggars sitting outside 10 gold coins and 5 diamonds.

1. How many gold coins and diamonds did they give away in all?
2. How many gold coins and diamonds are left?
3. If Satyanweshi and Rahul share what is left equally, how many coins and diamonds does each get?

ANSWER KEY (VOLUNTEERS ONLY)

- 1: 5 × 10 = **50** coins and 5 × 5 = **25** diamonds.
- 2: 140 − 50 = **90** coins; 80 − 25 = **55** diamonds.
- 3: 90 ÷ 2 = **45** coins each. 55 ÷ 2 = 27 with **1 left over**: each gets **27** diamonds and one is left. Ask: "What can the boys do with the extra diamond?" (Keep it for the temple, give it to someone, or share its worth.) This is a nice way to meet a **remainder**.

Student worksheet

Name: _____	Date: _____

১) নীচের অঙ্কগুলোর স্থানীয় মান বলো: ১২, ২৫, ৫৬, ১৫৬, ২৫৬৭, ৮৪৫৬১
1) Say the place value of each digit:

২) টাকায় ভেঙে ছকটা পূরণ করো (যেমন ২৫ টাকা = ২টা দশ টাকার নোট + ৫টা এক টাকার কয়েন):
2) Fill in the table (for example Rs 25 = two ten-rupee notes + five one-rupee coins):

কতো টাকা	কটা ১০০ টাকার নোট	কটা ১০ টাকার নোট	কটা ১ টাকার কয়েন
১২			
২৫			
২৫৬			

৩) বাক্সে ৩টি করে বিড়ালছানা। ৪টি বাক্সে মোট কতটি? ৩ + ৩ + ___ + ___ = ___। ___টা বাক্স × প্রতিটিতে ___টা ছানা = মোট ___টা ছানা
3) Each box has 3 kittens. How many in 4 boxes?

৪) ২৫ × ৩ = (২০ × ৩) + (৫ × ৩) = _____ + _____ = _____
4) 25 × 3 = (20 × 3) + (5 × 3) = _____ + _____ = _____

৫) এই নিয়মে আরও কয়েকটা গুণ করো:
5) More multiplications in the same way:

৩৫ × ৮ _____	৫৬ × ৬ _____	১০৯ × ৯ _____

৬) ১২টা বস্তা, প্রতি বস্তায় ১৪টা করে কুমড়ো। মোট কুমড়ো = ১৪ × ১২। কাঠি সাজিয়ে ছেদবিন্দু গুনে ছকটা ভরো:
6) 12 sacks, 14 pumpkins in each. Total = 14 × 12. Lay out sticks, count the crossings, and fill in the table:

অঞ্চল	ছেদবিন্দু	স্থানীয় মান
শতক		
দশক		
একক		
মোট		

৭) কাঠি সাজিয়ে আরও গুণ করো: ১৫ × ১১, ২১ × ২১, ১০ × ১২
7) More with sticks: 15 × 11, 21 × 21, 10 × 12

৮) ১৪ × ১২ = (১০ × ১২) + (৪ × ১২) = _____ + _____ = _____। এবার ১৫ × ২২ = _____
8) 14 × 12 = (10 × 12) + (4 × 12) = _____ + _____ = _____. Now 15 × 22 = _____

৯) গল্পের অঙ্ক: সত্যান্বেষী ও মন্দির
9) Story problem: Satyanweshi and the temple

(ক) মন্দির ৭ কিমি উপরে, ১ কিমি যেতে ১৫০০ পা লাগে। মোট কত পা?
(a) The temple is 7 km uphill, 1500 steps per km. How many steps?

(খ) ১৯টি কলা ৫ জনের মধ্যে সমান ভাগ করলে সত্যান্বেষীর কাছে কয়টি থাকবে?
(b) 19 bananas shared equally among 5. How many are left with Satyanweshi?

(গ) ৬০০০ পা হাঁটার পর আর কত পা বাকি?
(c) After 6000 steps, how many are left?

(ঘ) নামার সময় অর্ধেক পথ ১৫০০ পায়ে প্রতি কিমি, বাকি অর্ধেক ২০০০ পায়ে প্রতি কিমি। মোট কত পা?
(d) Coming down: half the way at 1500 steps per km, half at 2000. Total steps?

১০) ধাঁধা: আমি একটি তিন অঙ্কের সংখ্যা। আমার প্রথম অঙ্ক বিজোড়, ২-এর বেশি। দ্বিতীয় অঙ্ক ১ থেকে ১০-এর মধ্যে সবচেয়ে বড় জোড় সংখ্যা। তৃতীয় অঙ্ক দ্বিতীয়টির অর্ধেকের চেয়ে বড় এবং জোড়। তিন অঙ্কের যোগফল ১৭। আমি কত?
10) Riddle (see the volunteer page).

কোড: ___ ___ ___

১১) গুপ্তধন: ১৪০টি সোনার মুদ্রা ও ৮০টি হিরে। ৫ জন ভিখারিকে ১০টি করে মুদ্রা ও ৫টি করে হিরে দিলে: (ক) মোট কত দিল? (খ) কত বাকি? (গ) দুজনে সমান ভাগ করলে প্রত্যেকে কত পাবে?
11) Treasure: 140 gold coins, 80 diamonds. 5 beggars get 10 coins and 5 diamonds each: (a) how many given? (b) how many left? (c) how many each if two share equally?

	সোনার মুদ্রা	হিরে
(ক) মোট দিল		
(খ) বাকি		
(গ) প্রত্যেকে পাবে		